

# Daniel Fried

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|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONTACT<br>INFORMATION | dfried@cs.cmu.edu<br>dpfried.github.io                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                   |
| POSITIONS              | <b>Assistant Professor</b> , Carnegie Mellon University<br>Language Technologies Institute, School of Computer Science<br><b>Research Scientist</b> , Meta<br><i>Areas</i> : LLM-based coding agents<br>Currently 20% time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2022 – present<br><br>2024 – present                                                                                                                              |
| EDUCATION              | <b>UC Berkeley</b><br>Ph.D. in Computer Science<br><i>Adviser</i> : Dan Klein<br><i>Thesis</i> : Learning Grounded Pragmatic Communication<br><b>University of Cambridge</b> , Churchill College<br>M.Phil. in Computer Science, <i>with distinction</i><br><i>Adviser</i> : Stephen Clark<br><i>Thesis</i> : Low Rank Tensor Approximations for Compositional Distributional Semantics<br><b>University of Arizona</b><br>B.S. in Computer Science, Mathematics, and Information Science, <i>summa cum laude</i><br><i>Thesis Adviser</i> : Mihai Surdeanu<br><i>Thesis</i> : Predicting Community Traits Using the Language of Food on Social Media                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2015 – 2021<br><br><br><br>2014 – 2015<br><br><br><br>2010 – 2014                                                                                                 |
| FORMER<br>POSITIONS    | <b>Research Consultant</b> , GitHub Next<br><i>Areas</i> : LLM-based models for code editing<br><b>Postdoc</b> , University of Washington<br><i>Host</i> : Luke Zettlemoyer<br><b>Visiting Researcher</b> , Facebook AI Research<br><i>Host</i> : Mike Lewis<br><b>Research Intern</b> , Google DeepMind<br><i>Hosts</i> : Aida Nematzadeh, Stephen Clark, Chris Dyer<br><i>Project</i> : Structured models for language-conditioned video segmentation<br><b>Research Intern</b> , Microsoft Research<br><i>Hosts</i> : Hoifung Poon, Chris Quirk, Kristina Toutanova, Scott Wen-Tau Yih<br><i>Project</i> : Learning to rank for personalized medicine<br><b>Undergraduate Research Assistant</b> , University of Arizona<br><i>Advisers</i> : Mihai Surdeanu, Stephen Kobourov, Paul R. Cohen<br><i>Areas</i> : Question answering, language grounding, data visualization<br><b>Research Intern</b> , Nara Institute of Science and Technology<br><i>Host</i> : Kevin Duh<br><i>Project</i> : Incorporating relational semantics in word embeddings<br><b>Research Intern</b> , RWTH Aachen University<br><i>Hosts</i> : Ali Jannesari, Zhen Li<br><i>Project</i> : Supervised learning for automatic code parallelization<br><b>Engineering Intern</b> , Microsoft<br><i>Project</i> : Data warehousing, analytics, and visualization for Xbox Live | 2023 – 2024<br><br>2021 – 2022<br><br>2021 – 2022<br><br>Summer 2019<br><br>Summer 2016<br><br>2012 – 2014<br><br>Fall 2013<br><br>Summer 2013<br><br>Summer 2012 |

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|--------------------------------|-----------------------------------------------------------------------|------------------------|
| <b>HONORS &amp;<br/>AWARDS</b> | Schmidt Sciences AI 2050 Early Career Fellow                          | 2026                   |
|                                | NSF CAREER Award                                                      | 2026                   |
|                                | Microsoft Research Faculty Fellowship                                 | 2026                   |
|                                | Okawa Research Award                                                  | 2023                   |
|                                | Google Ph.D. Fellowship in Natural Language Processing                | 2019 – 2021            |
|                                | Outstanding Graduate Student Instructor, UC Berkeley                  | 2018                   |
|                                | Outstanding Reviewer, ACL                                             | 2018, 2020, 2021, 2022 |
|                                | Outstanding Reviewer, NeurIPS                                         | 2019                   |
|                                | Best M.Phil. Student Award, Cambridge Computer Laboratory             | 2015                   |
|                                | Churchill Scholarship                                                 | 2014 – 2015            |
|                                | NDSEG Fellowship                                                      | 2014                   |
|                                | Finalist, Hertz Graduate Fellowship                                   | 2014                   |
|                                | Outstanding Senior Award in Research, U. Arizona College of Science   | 2014                   |
|                                | Outstanding Senior Award in Academics, U. Arizona Computer Science    | 2014                   |
|                                | Outstanding Senior Award in Academics, U. Arizona Information Science | 2014                   |
|                                | Barry M. Goldwater Scholarship                                        | 2013                   |
|                                | National Merit Scholar                                                | 2010 – 2014            |
|                                | Flinn Scholarship, Flinn Foundation of Arizona                        | 2010 – 2014            |
|                                | Presidential Scholar, U.S. Department of Education                    | 2010                   |

## JOURNAL PUBLICATIONS

- Tree Search for Language Model Agents**  
Jing Yu Koh, Stephen McAleer, Daniel Fried, and Ruslan Salakhutdinov  
*Transactions on Machine Learning Research (TMLR)*, 2025
- StarCoder: May the Source Be With You!**  
Raymond Li et al. (68 authors from the BigCode Project)  
*Transactions on Machine Learning Research (TMLR)*, 2023
- Human-Level Play in the Game of Diplomacy by Combining Language Models with Strategic Reasoning**  
FAIR Diplomacy Team  
*Science*, 2022
- Syntactic Structure Distillation Pretraining for Bidirectional Encoders**  
Adhiguna Kuncoro\*, Lingpeng Kong\*, Daniel Fried\*, Dani Yogatama, Laura Rimell, Chris Dyer, and Phil Blunsom  
*Transactions of the Association for Computational Linguistics (TACL)*, 2020
- Higher-Order Lexical Semantic Models for Non-Factoid Answer Reranking**  
Daniel Fried, Peter Jansen, Gustave Hahn-Powell, Mihai Surdeanu, and Peter Clark  
*Transactions of the Association for Computational Linguistics (TACL)*, 2015

## CONFERENCE PUBLICATIONS

- Reasoning with Latent Tokens in Diffusion Language Models**  
Andre He, Sean Welleck\*, Daniel Fried\*  
*NeurIPS*, 2026. **Spotlight (top 1% of submissions)**
- Odysseys: Benchmarking Web Agents on Realistic Long Horizon Tasks**  
Lawrence Jang\*, Jing Yu Koh\*, Daniel Fried, Ruslan Salakhutdinov  
*Conference on Language Modeling (COLM)*, 2026
- MetaLint: Generalizable Idiomatic Code Quality Analysis through Instruction-Following and Easy-to-Hard Generalization**  
Atharva Naik, Lawanya Baghel, Dhakshin Govindarajan, Darsh Agrawal, Yiqing Xie, Daniel Fried, Carolyn Rose  
*Conference on Language Modeling (COLM)*, 2026
- Agent Psychometrics: Task-Level Performance Prediction in Agentic Coding Benchmarks**  
Chris Ge, Daria Kryvosheieva, Daniel Fried, Uzay Girit, Kaivalya Hariharan  
*Conference on Language Modeling (COLM)*, 2026

10. **Scaling Test-Time Compute for Agentic Coding**  
Joongwon Kim, Wannan Yang, Kelvin Niu, Hongming Zhang, Yun Zhu, Eryk Helenowski, Ruan Silva, Zhengxing Chen, Srinivasan Iyer, Manzil Zaheer, Daniel Fried, Hannaneh Hajishirzi, Sanjeev Arora, Gabriel Synnaeve, Ruslan Salakhutdinov, Anirudh Goyal  
*Conference on Language Modeling (COLM)*, 2026
11. **Hybrid-Gym: Training Coding Agents to Generalize Across Tasks**  
Yiqing Xie, Emmy Liu, Gaokai Zhang, Nachiket Kotalwar, Shubham Gandhi, Sathwik Acharya, Xingyao Wang, Carolyn Rose, Graham Neubig, Daniel Fried  
*International Conference on Machine Learning (ICML)*, 2026
12. **Propose, Solve, Verify: Self-Play Through Formal Verification**  
Alex Wilf, Pranjal Aggarwal, Bryan Parno, Daniel Fried, Louis-Philippe Morency, Paul Pu Liang, Sean Welleck  
*International Conference on Machine Learning (ICML)*, 2026
13. **Toward Training Superintelligent Software Agents through Self-Play SWE-RL**  
Yuxiang Wei, Zhiqing Sun, Emily McMilin, Jonas Gehring, David Zhang, Gabriel Synnaeve, Daniel Fried, Lingming Zhang, Sida Wang  
*International Conference on Machine Learning (ICML)*, 2026
14. **Success and Cost Elicit Convention Formation for Efficient Communication**  
Saujas Vaduguru, Yilun Hua, Yoav Artzi, Daniel Fried  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026
15. **Generative Value Conflicts Reveal LLM Priorities**  
Andy Liu, Kshitish Ghate, Mona Diab\*, Daniel Fried\*, Atoosa Kasirzadeh\*, Max Kleiman-Weiner\*  
*International Conference on Learning Representations (ICLR)*, 2026
16. **From Reproduction to Replication: Evaluating Research Agents with Progressive Code Masking**  
Gyeongwon James Kim, Alex Wilf, Louis-Philippe Morency, Daniel Fried  
*International Conference on Learning Representations (ICLR)*, 2026
17. **SWE-RL: Advancing LLM Reasoning via Reinforcement Learning on Open Software Evolution**  
Yuxiang Wei, Olivier Duchenne, Jade Copet, Quentin Carbonneaux, Lingming Zhang, Daniel Fried, Gabriel Synnaeve, Rishabh Singh, and Sida I. Wang  
*Neural Information Processing Systems (NeurIPS)*, 2025
18. **Identifying and Interactively Refining Ambiguous User Goals for Data Visualization Code Generation**  
Mert Inan, Anthony Sicilia, Alex Xie, Saujas Vaduguru, Daniel Fried, and Malihe Alikhani  
*Empirical Methods in Natural Language Processing (EMNLP)*, 2025
19. **Rewarding the Unlikely: Lifting GRPO Beyond Distribution Sharpening**  
Andre He, Daniel Fried, and Sean Welleck  
*Empirical Methods in Natural Language Processing (EMNLP)*, 2025
20. **mrCAD: Multimodal Refinement of Computer-aided Designs**  
William P. McCarthy, Saujas Vaduguru, Karl D. D. Willis, Justin Matejka, Judith E. Fan, Daniel Fried, and Yewen Pu  
*Findings of EMNLP*, 2025
21. **Inducing Programmatic Skills for Agentic Tasks**  
Zora Zhiruo Wang, Apurva Gandhi, Graham Neubig, Daniel Fried  
*Conference on Language Modeling (COLM)*, 2025
22. **RepoST: Scalable Repository-Level Coding Environment Construction with Sandbox Testing**  
Yiqing Xie, Alex Xie, Divyanshu Sheth, Pengfei Liu, Daniel Fried, Carolyn Rose  
*Conference on Language Modeling (COLM)*, 2025
23. **Improving Model Factuality with Fine-grained Critique-based Evaluator**  
Yiqing Xie, Wenxuan Zhou, Pradyot Prakash, Di Jin, Yuning Mao, Quintin Fettes, Arya Talebzadeh, Sinong Wang, Han Fang, Carolyn Rose, Daniel Fried, and Hejia Zhang  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025

24. **Agent Workflow Memory**  
Zora Zhiruo Wang, Jiayuan Mao, [Daniel Fried](#), and Graham Neubig  
*ICML*, 2025
25. **Dynamic Coalition Structure Detection in Natural Language-based Interactions**  
Abhishek N. Kulkarni\*, Andy Liu\*, Jean-Raphael Gaglione, [Daniel Fried](#), and Ufuk Topcu  
*International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2025
26. **AutoPresent: Designing Structured Visuals from Scratch**  
Jiaxin Ge\*, Zora Zhiruo Wang\*, Xuhui Zhou, Yi-Hao Peng, Sanjay Subramanian, Qinyue Tan, Maarten Sap, Alane Suhr\*\*, [Daniel Fried](#)\*\*, Graham Neubig\*\*, and Trevor Darrell\*\*  
*Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025
27. **CRScore: Grounding Automated Evaluation of Code Review Comments in Code Claims and Smells**  
Atharva Naik, Marcus Alenius, [Daniel Fried](#), and Carolyn Rose  
*North American Chapter of the Association for Computational Linguistics (NAACL)*, 2025
28. **CodeRAG-Bench: Can Retrieval Augment Code Generation?**  
Zora Zhiruo Wang\*, Akari Asai\*, Xinyan Velocity Yu, Frank F. Xu, Yiqing Xie, Graham Neubig, and [Daniel Fried](#)  
*Findings of NAACL*, 2025
29. **BigCodeBench: Benchmarking Code Generation with Diverse Function Calls and Complex Instructions**  
Terry Yue Zhuo et al. (33 authors from the BigCode project)  
*International Conference on Learning Representations (ICLR)*, 2025
30. **Human-Aligned Chess with a Bit of Search**  
Yiming Zhang, Athul Paul Jacob, Vivian Lai, [Daniel Fried](#), and Daphne Ippolito  
*International Conference on Learning Representations (ICLR)*, 2025
31. **Repetition Improves Language Model Embeddings**  
Jacob Mitchell Springer, Suhas Kotha, [Daniel Fried](#), Graham Neubig, and Aditi Raghunathan  
*International Conference on Learning Representations (ICLR)*, 2025
32. **Dissecting Adversarial Robustness of Multimodal LM Agents**  
Chen Henry Wu, Rishi Shah, Jing Yu Koh, Ruslan Salakhutdinov, [Daniel Fried](#), and Aditi Raghunathan  
*International Conference on Learning Representations (ICLR)*, 2025
33. **Comparative Knowledge Distillation**  
Alex Tianyi Xu\*, Alex Wilf\*, Paul Pu Liang, Alexander Obolenskiy, [Daniel Fried](#), and Louis-Philippe Morency  
*Winter Conference on Applications of Computer Vision (WACV)*, 2024
34. **ECCO: Can We Improve Model-Generated Code Efficiency Without Sacrificing Functional Correctness?**  
Siddhant Waghjale\*, Vishruth Veerendranath\*, Zora Zhiruo Wang, and [Daniel Fried](#)  
*Empirical Methods in Natural Language Processing (EMNLP)*, 2024
35. **What Are Tools Anyway? A Survey from the Language Model Perspective**  
Zora Zhiruo Wang, Zhoujun Cheng, Hao Zhu, [Daniel Fried](#), and Graham Neubig  
*Conference on Language Modeling (COLM)*, 2024
36. **Human-Agent Cooperation in Games under Incomplete Information through Natural Language Communication**  
Shenghui Chen, [Daniel Fried](#), and Ufuk Topcu  
*International Joint Conference on Artificial Intelligence (IJCAI)*, 2024
37. **Evaluating Large Language Model Biases in Persona-Steered Generation**  
Andy Liu, Mona T. Diab, and [Daniel Fried](#)  
*Findings of ACL*, 2024
38. **Is the Pope Catholic? Yes, the Pope is Catholic. Generative Evaluation of Intent Resolution in LLMs**

- Akhila Yerukola, Saujas Vaduguru, [Daniel Fried](#), and Maarten Sap  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024
39. **VisualWebArena: Evaluating Multimodal Agents on Realistic Visual Web Tasks**  
Jing Yu Koh, Robert Lo\*, Lawrence Jang\*, Vikram Duvvur\*, Ming Chong Lim\*, Po-Yu Huang\*, Graham Neubig, Shuyan Zhou, Ruslan Salakhutdinov, and [Daniel Fried](#)  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024
40. **TroVE: Inducing Verifiable and Efficient Toolboxes for Solving Programmatic Tasks**  
Zhiruo Wang, Graham Neubig, and [Daniel Fried](#)  
*International Conference on Machine Learning (ICML)*, 2024
41. **Amortizing Pragmatic Program Synthesis with Rankings**  
Yewen Pu, Saujas Vaduguru, Priyan Vaithilingam, Elena Glassman, and [Daniel Fried](#)  
*International Conference on Machine Learning (ICML)*, 2024
42. **Asking More Informative Questions for Grounded Retrieval**  
Sedrick Keh, Justin T. Chiu, and [Daniel Fried](#)  
*Findings of NAACL*, 2024
43. **Generating Pragmatic Examples to Train Neural Program Synthesizers**  
Saujas Vaduguru, [Daniel Fried](#), and Yewen Pu  
*International Conference on Learning Representations (ICLR)*, 2024
44. **Sotopia: Interactive Evaluation for Social Intelligence in Language Agents**  
Xuhui Zhou\*, Hao Zhu\*, Leena Mathur, Ruohong Zhang, Haoifei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, [Daniel Fried](#), Graham Neubig, and Maarten Sap  
*International Conference on Learning Representations (ICLR)*, 2024
45. **WebArena: A Realistic Web Environment for Building Autonomous Agents**  
Shuyan Zhou\*, Frank Xu\*, Hao Zhu\*\*, Xuhui Zhou\*\*, Robert Lo\*\*, Abishek Sridhar\*\*, Xianyi Cheng, Yonatan Bisk, [Daniel Fried](#), Uri Alon, and Graham Neubig  
*International Conference on Learning Representations (ICLR)*, 2024
46. **API-Assisted Code Generation for Question Answering on Varied Table Structures**  
Yihan Cao\*, Shuyi Chen\*, Ryan Liu\*, Zhiruo Wang, and [Daniel Fried](#)  
*Empirical Methods in Natural Language Processing (EMNLP)*, 2023
47. **Symbolic Planning and Code Generation for Grounded Dialogue**  
Justin Chiu, Wenting Zhao, Derek Chen, Saujas Vaduguru, Alexander Rush, and [Daniel Fried](#)  
*Empirical Methods in Natural Language Processing (EMNLP)*, 2023
48. **Pragmatics in Language Grounding: Phenomena, Tasks, and Modeling Approaches**  
[Daniel Fried\\*](#), Nicholas Tomlin\*, Jennifer Hu, Roma Patel, and Aida Nematzadeh  
*Findings of EMNLP*, 2023
49. **Execution-Based Evaluation for Open-Domain Code Generation**  
Zhiruo Wang, Shuyan Zhou, [Daniel Fried](#), and Graham Neubig  
*Findings of EMNLP*, 2023
50. **Data Augmentation for Code Translation with Comparable Corpora and Multiple References**  
Yiqing Xie, Atharva Naik, [Daniel Fried](#), Carolyn Rose  
*Findings of EMNLP*, 2023
51. **AutoReply: Detecting Nonsense in Dialogue Introspectively with Discriminative Replies**  
Weiyan Shi, Emily Dinan, Adi Renduchintala, [Daniel Fried](#), Athul Paul Jacob, Zhou Yu, and Mike Lewis  
*Findings of EMNLP*, 2023
52. **Generating Images with Multimodal Language Models**  
Jing Yu Koh, [Daniel Fried](#), and Ruslan Salakhutdinov  
*Neural Information Processing Systems (NeurIPS)*, 2023
53. **Pragmatic Inference with a CLIP Listener for Contrastive Captioning**  
Jiefu Ou, Benno Krojer, and [Daniel Fried](#)  
*Findings of ACL*, 2023

54. **Contrastive Decoding: Open-ended Text Generation as Optimization**  
Xiang Lisa Li, Ari Holtzman, Daniel Fried, Percy Liang, Jason Eisner, Tatsunori Hashimoto, Luke Zettlemoyer, and Mike Lewis  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023
55. **Grounding Language Models to Images for Multimodal Inputs and Outputs**  
Jing Yu Koh, Ruslan Salakhutdinov, and Daniel Fried  
*International Conference on Machine Learning (ICML)*, 2023
56. **Coder Reviewer Reranking for Code Generation**  
Tianyi Zhang, Tao Yu, Tatsunori B. Hashimoto, Mike Lewis, Wen-tau Yih, Daniel Fried, and Sida I. Wang  
*International Conference on Machine Learning (ICML)*, 2023
57. **DS-1000: A Natural and Reliable Benchmark for Data Science Code Generation**  
Yuhang Lai\*, Chengxi Li\*, Yiming Wang\*, Tianyi Zhang\*, Ruiqi Zhong\*, Luke Zettlemoyer, Scott Wen-tau Yih, Daniel Fried, Sida I. Wang, and Tao Yu  
*International Conference on Machine Learning (ICML)*, 2023
58. **InCoder: A Generative Model for Code Infilling and Synthesis**  
Daniel Fried\*, Armen Aghajanyan\*, Jessy Lin, Sida I. Wang, Eric Wallace, Freda Shi, Ruiqi Zhong, Wen-tau Yih, Luke Zettlemoyer, and Mike Lewis  
*International Conference on Learning Representations (ICLR)*, 2023
59. **Natural Language to Code Translation with Execution**  
Freda Shi, Daniel Fried, Marjan Ghazvininejad, Luke Zettlemoyer, and Sida I. Wang  
*Empirical Methods in Natural Language Processing (EMNLP)*, 2022
60. **Neural Theory-of-Mind? On the Limits of Social Intelligence in Large LMs**  
Maarten Sap, Ronan Le Bras, Daniel Fried, and Yejin Choi  
*Empirical Methods in Natural Language Processing (EMNLP)*, 2022
61. **G3: Geolocation via Guidebook Grounding**  
Grace Luo\*, Giscard Biamby\*, Trevor Darrell, Daniel Fried, and Anna Rohrbach  
*Findings of EMNLP*, 2022
62. **Inferring Rewards from Language in Context**  
Jessy Lin, Daniel Fried, Dan Klein, and Anca Dragan  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022
63. **Reference-Centric Models for Grounded Collaborative Dialogue**  
Daniel Fried, Justin Chiu, and Dan Klein  
*Empirical Methods in Natural Language Processing (EMNLP)*, 2021
64. **Modular Networks for Compositional Instruction Following**  
Rodolfo Corona, Daniel Fried, Coline Devin, Dan Klein, and Trevor Darrell  
*North American Chapter of the Association for Computational Linguistics (NAACL)*, 2021
65. **Learning to Segment Actions from Observation and Narration**  
Daniel Fried, Jean-Baptiste Alayrac, Phil Blunsom, Chris Dyer, Stephen Clark, Aida Nematzadeh  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020
66. **Cross-Domain Generalization of Neural Constituency Parsers**  
Daniel Fried\*, Nikita Kitaev\*, and Dan Klein  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019
67. **Are You Looking? Grounding to Multiple Modalities in Vision-and-Language Navigation**  
Ronghang Hu, Daniel Fried, Anna Rohrbach, Dan Klein, Trevor Darrell, and Kate Saenko  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019
68. **Pragmatically Informative Text Generation**  
Sheng Shen, Daniel Fried, Jacob Andreas, and Dan Klein  
*North American Chapter of the Association for Computational Linguistics (NAACL)*, 2019
69. **Speaker-Follower Models for Vision-and-Language Navigation**  
Daniel Fried\*, Ronghang Hu\*, Volkan Cirik\*, Anna Rohrbach, Jacob Andreas, Louis-Philippe Morency,

- Taylor Berg-Kirkpatrick, Kate Saenko, Dan Klein\*\*, and Trevor Darrell\*\*  
*Neural Information Processing Systems (NeurIPS)*, 2018
70. **Policy Gradient as a Proxy for Dynamic Oracles in Constituency Parsing**  
Daniel Fried and Dan Klein  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018
  71. **Unified Pragmatic Models for Generating and Following Instructions**  
Daniel Fried, Jacob Andreas, and Dan Klein  
*North American Chapter of the Association for Computational Linguistics (NAACL)*, 2018
  72. **Effective Inference for Generative Neural Parsing**  
 Mitchell Stern, Daniel Fried, and Dan Klein  
*Empirical Methods in Natural Language Processing (EMNLP)*, 2017
  73. **Improving Neural Parsing by Disentangling Model Combination and Reranking Effects**  
Daniel Fried\*, Mitchell Stern\*, and Dan Klein  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2017
  74. **Towards Using Social Media to Identify Individuals at Risk for Preventable Chronic Illness**  
 Dane Bell, Daniel Fried, Luwen Huangfu, Mihai Surdeanu, and Stephen Kobourov  
*Language Resources and Evaluation Conference (LREC)*, 2016
  75. **Low-Rank Tensors for Verbs in Compositional Distributional Semantics**  
Daniel Fried, Tamara Polajnar, and Stephen Clark  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2015
  76. **Analyzing the Language of Food on Social Media**  
Daniel Fried, Mihai Surdeanu, Stephen Kobourov, Melanie Hingle, and Dane Bell  
*International Conference on Big Data*, 2014
  77. **Maps of Computer Science**  
Daniel Fried and Stephen Kobourov  
*Pacific Visualization Symposium (PacificVis)*, 2014
  78. **Predicting Parallelization of Sequential Programs Using Supervised Learning**  
Daniel Fried, Zhen Li, Ali Jannesari, and Felix Wolf  
*International Conference on Machine Learning and Applications*, 2013
  79. **A Generative Probabilistic Framework for Learning Spatial Language**  
 Colin Dawson, Jeremy Wright, Antons Rebguns, Marco Valenzuela Escarcega, Daniel Fried, and Paul Cohen  
*International Conference on Development and Learning*, 2013. **Best Paper Award**
  80. **Bayesian Geometric Modeling of Indoor Scenes**  
 Luca Del Pero, Joshua Bowdish, Daniel Fried, Bonnie Kermgard, Emily Hartley, and Kobus Barnard  
*Conference on Computer Vision and Pattern Recognition (CVPR)*, 2012

#### WORKSHOP PUBLICATIONS

81. **SantaCoder: Don't Reach for the Stars**  
 Loubna Ben Allal\*, Raymond Li\*, Denis Kocetkov\*, et al. (41 authors from the BigCode Project)  
*Deep Learning for Code Workshop*, 2023. **Best Paper Award**
82. **Modeling Perspective-Dependent Ambiguity in Grounded Collaborative Dialogue**  
 Justin Chiu, Wenting Zhao, Alexander M. Rush, and Daniel Fried  
*Wordplay: When Language Meets Games Workshop*, 2022
83. **Interactive Assignments for Teaching Structured Neural NLP**  
 David Gaddy, Daniel Fried, Nikita Kitaev, Mitchell Stern, Rodolfo Corona, John DeNero, and Dan Klein  
*Teaching NLP Workshop at NAACL*, 2021
84. **Challenges for Using Social Media for Early Detection of Type II Diabetes Mellitus**  
 Dane Bell, Daniel Fried, Luwen Huangfu, Mihai Surdeanu, and Stephen Kobourov  
*International Workshop on Social Media World Sensors*, 2016

85. **Learning Low-Rank Tensors for Transitive Verbs**  
Daniel Fried, Tamara Polajnar, and Stephen Clark  
*Advances in Distributional Semantics Workshop*, 2015
86. **Incorporating both Distributional and Relational Semantics in Word Representations**  
Daniel Fried and Kevin Duh  
*International Conference on Learning Representations (ICLR) Workshop*, 2015

#### PREPRINTS

87. **CUA-Speedrun: Standardized Benchmarking of the Speed of Computer-Use Agents**  
Pranjal Aggarwal\*, Lawrence Keunho Jang\*, Sean Welleck, Daniel Fried, Ruslan Salakhutdinov, Jing Yu Koh\*  
*preprint*, 2026
88. **Efficient Test-Time Adaptation through Human-AI Interaction**  
Zora Zhiruo Wang, Apurva Gandhi, Rulin Shao, Aspen Chen, Jonas Mueller, Zhiqi Liang, Jett Chen, Michael Ryan, Qianou Ma, Luxi He, Zhoujun Cheng, Andre He, Seungone Kim, Jiayi Geng, Mingqian Zheng, Weiwei Sun, Zheyuan Zhang, Xinran Zhao, Yike Wang, Abe Hou, Liwei Jiang, Pang Wei Koh, Diyi Yang, Graham Neubig, Daniel Fried  
*arXiv*, 2026
89. **Multi-Agent Computer Use**  
Jing Yu Koh, Ruslan Salakhutdinov, Daniel Fried  
*arXiv*, 2026
90. **How Well Does Agent Development Reflect Real-World Work?**  
Zora Zhiruo Wang, Sanidhya Vijayvargiya, Aspen Chen, Hanmo Zhang, Venu Arvind Arangarajan, Jett Chen, Valerie Chen, Diyi Yang, Daniel Fried, Graham Neubig  
*arXiv*, 2026
91. **Position: Humans are Missing from AI Coding Agent Research**  
Zora Zhiruo Wang\*, John Yang\*, Kilian Lieret\*, Alexa Tartaglino, Valerie Chen, Yuxiang Wei, Zijian Wang, Lingming Zhang, Karthik Narasimhan, Ludwig Schmidt, Graham Neubig, Daniel Fried, Diyi Yang  
*preprint*, 2026
92. **Measuring Fine-Grained Negotiation Tactics of Humans and LLMs in Diplomacy**  
Wenkai Li\*, Lynnette Hui Xian Ng\*, Andy Liu, Daniel Fried  
*arXiv*, 2025
93. **How Do AI Agents Do Human Work? Comparing AI and Human Workflows Across Diverse Occupations**  
Zora Zhiruo Wang, Yijia Shao, Omar Shaikh, Daniel Fried, Graham Neubig, Diyi Yang  
*arXiv*, 2025
94. **Analyzing Information Sharing and Coordination in Multi-Agent Planning**  
Tianyu Ou, Saujas Vaduguru, Daniel Fried  
*arXiv*, 2025
95. **CodeBenchGen: Creating Scalable Execution-based Code Generation Benchmarks**  
Yiqing Xie, Alex Xie, Divyanshu Sheth, Pengfei Liu, Daniel Fried, and Carolyn Rose  
*arXiv*, 2024

\*, \*\*: equal contribution

#### STUDENTS

PhD thesis advisor:

- Andre He (with Sean Welleck)
- Jing Yu (JY) Koh (with Ruslan Salakhutdinov)
- Andy Liu (with Mona Diab)
- Saujas Vaduguru
- Zhiruo (Zora) Wang (with Graham Neubig)

MLT thesis advisor:

- Alex Xie (with Matt Gormley and Vincent Hellendoorn)

PhD committee member:

- Shuyan Zhou
- Aman Madaan
- Ta-Chung Chi
- Nikitha Rao
- Frank Xu
- Ritam Dutt
- Ruohong Zhang
- So Yeon (Tiffany) Min
- Anthony Sicilia (Northeastern)
- Kush Jain
- Brendon Boldt
- Nicholas Tomlin (UC Berkeley)
- Sanjay Subramanian (UC Berkeley)
- Liangke Gui
- Xing Han Liu (McGill)
- Xuhui Zhou
- Yiqing Xie
- Alex Wilf
- Yuxiang Wei (UIUC)
- Akhila Yerukola

#### TEACHING

***Instructor: 11-768, AI Agents, CMU, 2026 – present***

***Instructor: 11-891, Neural Code Generation, CMU, 2024 – present***

***Instructor: 11-777, Multimodal Machine Learning, CMU, 2023 – present***

***Instructor: 11-877, Advanced Multimodal Machine Learning, CMU, Spring 2024***

***Instructor: 11-711, Advanced NLP, CMU, Fall 2023***

***Instructor: CS188, Artificial Intelligence, UC Berkeley, Summer 2018***

***Teaching Assistant: CS188, Artificial Intelligence, UC Berkeley, Fall 2017***

***Teaching Assistant: CS245, Intro to Discrete Structures, University of Arizona, Spring 2012***

***Teaching Assistant: ISTA100, Great Ideas of the Information Age, University of Arizona, Fall 2011***

***Project Design: CS288, Natural Language Processing, UC Berkeley, Spring 2020***

**INVITED TALKS** *Doing More With Less: Behavioral Efficiency in Language Model Agents*. Stanford NLP Seminar. Spring 2026

*Abstractions of Agent Actions*. IVADO Workshop on the Safety and Capabilities of Agents 2025; Agentic AI Frontier Seminar 2026; CMU HCII Seminar 2026; UPenn NLP Seminar 2026; MIT NLP Seminar 2026

*Improving NLP Systems with Speaker Listener Games*. Pragmatic Reasoning in Language Models Workshop, COLM 2025.

*Code Generation With and For Agents*. IVADO Bootcamp Focusing the State of Agents 2025.

*Improving NLP Systems with Speaker Listener Games*. **Keynote Address**, Society for Computation in Linguistics (SCiL) 2025.

*Inducing Functions to Improve LLM Agents*. Deep Learning For Code Workshop, ICLR 2025; REALM Workshop, ACL 2025.

*Planning and Inferring with LLMs for Grounded, Interactive Tasks*. Forum for Artificial Intelligence, UT Austin. Fall 2024

*Benchmarks and Tree Search for Multimodal LLM Agents*. SpLU-RoboNLP Workshop, ACL 2024. Summer 2024

*Interacting with LLMs for Grounded Tasks*. NILLI Workshop, EMNLP 2023. Spring 2023

*Interacting with (code) LLMs*. MIT Neurosymbolic Reading Group. Spring 2023

*InCoder, SantaCoder, and StarCoder: Findings from Training Open-Source Code LLMs*. Bloomberg AI; GitHub Next. Spring 2023

*Using Language Strategically in Context*. CoRL Workshop on Strategic Multi-Agent Interactions, 2022; Johns Hopkins University, University of Pennsylvania, UT Austin. Spring 2023

*Contextual Communication in Programming*. LTI Future of Code Generation Seminar, 2022

*Reasoning About Actions and Rewards in Language Interactions*. MIT CPL, 2022. (with Jessy Lin)

*Modularity in Grounded Interaction*. ViGIL Workshop, NAACL 2021. (with Rudy Corona)

*Learning Grounded Pragmatic Communication*. University of Arizona, TTI-Chicago, University of Southern California, Purdue, Carnegie Mellon University, UC Irvine, Université de Montréal/Mila, Allen Institute for Artificial Intelligence, Facebook AI Research, Google Research, Stanford Cognition and Language Workshop. Spring 2021

*Pragmatic Models for Generating and Following Grounded Instructions*. Stanford NLP Seminar, University of Arizona, USC ISI. Fall 2018–Spring 2019

**ACADEMIC  
SERVICE**

**Workshop co-organizer:** Workshop on Large Language Models for Agents, ICLR 2024

**Workshop co-organizer:** 3rd UnImplicit Workshop, EACL 2024

**Workshop co-organizer:** 2nd UnImplicit Workshop, NAACL 2022

**Workshop co-organizer:** 2nd Advances in Language and Vision Workshop, NAACL 2021

**Workshop advisory board:** Theory of Mind in Communicating Agents, ICML 2023

**Outstanding reviewer awards:** ACL 2018, NeurIPS 2019, ACL 2020, ACL 2021, ACL 2022

**Senior area chair:** ACL 2024; ACL 2026

**Area chair:** EMNLP 2022–2024, ACL 2023, ACL Rolling Review 2024–present

**Ethics chair:** NAACL 2024

**Action Editor:** TACL 2025–present

**Reviewing:** TACL 2022–2025; ACL Rolling Review 2021–2024; ACL 2018–2022; EMNLP 2016–2021; NAACL 2019, 2021; AACL-IJCNLP 2020; EACL 2017, 2021; NeurIPS 2019–2023; ICML 2019, 2023, 2026; ICLR 2021–2023; AKBC 2021; COLING 2018; \*SEM 2016–2018; NAACL-SRW 2016, 2018; ACL-SRW 2020; IJCAI 2016; SpLU-RoboNLP 2019, 2021; NeuralGen 2019; ViGIL 2019, 2021; DeepLo 2019; ALVR 2020, 2021; LaReL 2020; Cooperative AI 2021; Meaning in Context 2021